SERVICE MANUAL

MRI Devices Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0103S

Revision A

GE Signa® 1.0T / 1.5T
Open Neurovascular Array

GE Catalog Part Number:	**M1887NB**	**(1.0T)**
	M1087NB	**(1.5T)**
MRIDC Part Number:	**800201**	**(1.0T)**
	800202	**(1.5T)**

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF

Rev. 11/15/2000

Language Policy For Service Documentation (Dir. 2128126)

W A R N I N G

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON–RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇAO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルは英語版しかありません。

・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。

・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

TABLE OF CONTENTS

SECTION 1 - INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE Signa 1.0T / 1.5T Open Neurovascular Array.

1.0T SHIPPING LIST – TABLE 1-1-1

Description	GE Part #	MRIDC Part #	Qty
Neurovascular Assembly with Cable	2293669-2	101423	1
Neurovascular Operator Manual	2293668-3	500067	1
Neurovascular Service Manual	2293668-4	TR0103S	1
Wedge Pad	E8800NB	101407	2
Pad	E8800NC	101408	1

1.5T SHIPPING LIST – TABLE 1-1-2

Description	GE Part #	MRIDC Part #	Qty
Neurovascular Assembly with Cable	2293668-2	101413	1
Neurovascular Operator Manual	2293668-3	500067	1
Neurovascular Service Manual	2293668-4	TR0103S	1
Wedge Pad	E8800NB	101407	2
Pad	E8800NC	101408	1

1-2 Compatibility

This coil is compatible with the following hardware configurations:

- M1887NB - 1.0T Signa LX/Horizon, 1.0T Signa Cardiac, and 1.0T Signa Advantage
- M1087NB - 1.5T Signa LX/Horizon, 1.5T Signa Cardiac, and 1.5T Signa Advantage

1-3 Related Documentation

Signa LX Service Methods CD, 2160623-1

1.0T/1.5T Open Neurovascular Array Operator Manual, 2293668-3

1-4 Environmental Requirements

Storage Requirements

Coil should be stored in the Scanner Room.

Dimensions

Coil Dimension:	679.5 mm x 394.0 mm x 415.9 mm	(26.75 in x 15.50 in x 16.375 in)

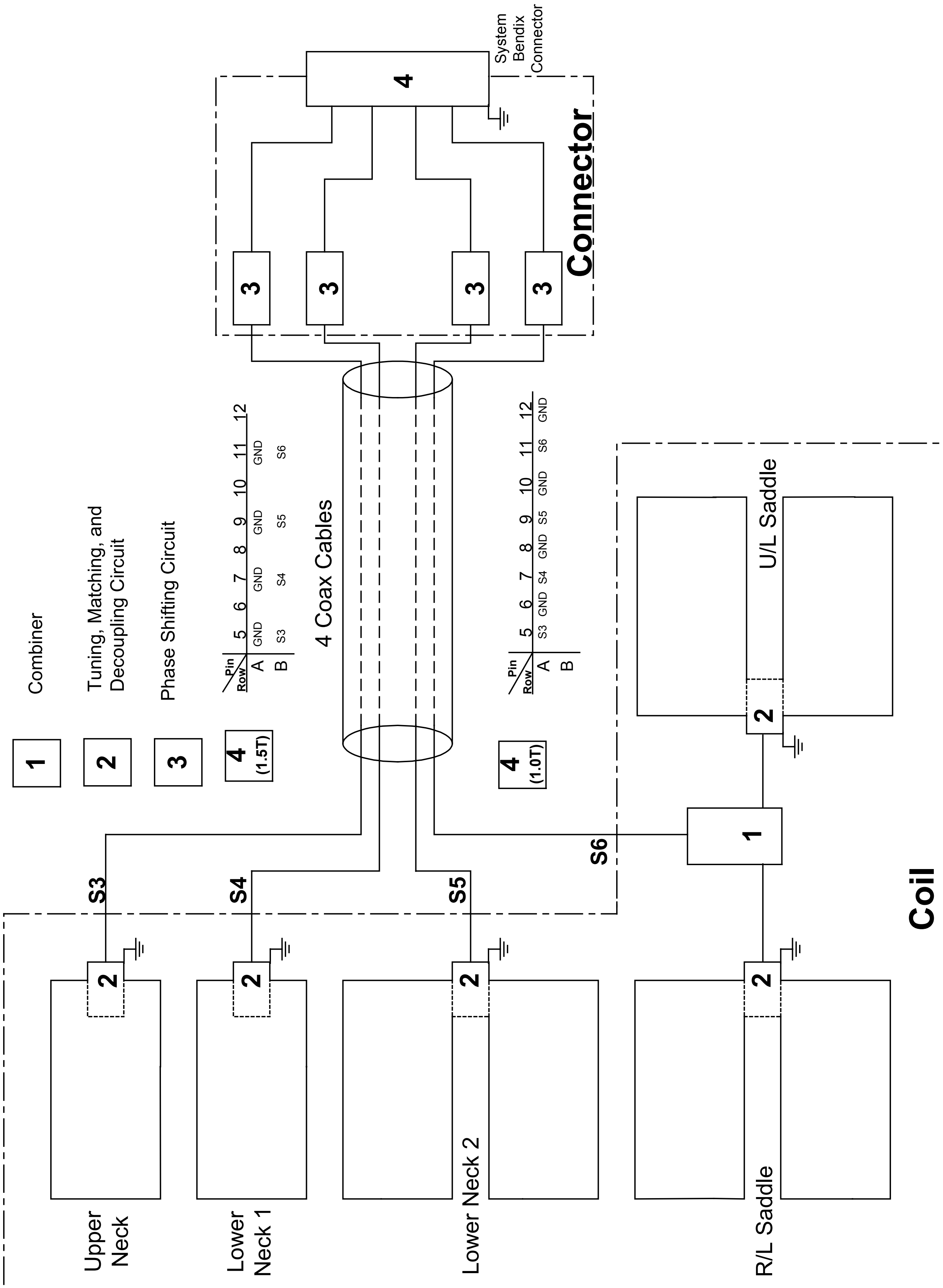
Weight

Coil Weight:	9.77 Kg	(21.5 lb.)

1-5 Theory of Operation

Refer to the block diagram on the following page.

The Phased Array Neurovascular coil consists of a top and bottom piece which must be securely fastened together for safe effective operation. The coil utilizes four channels of the multi-coil system. The head channel on S6 is a quadrature saddle coil which uses the combiner labeled (1). The neck coverage comes from three channels, one for the anterior neck and the other two for the posterior neck. The coils are tuned/matched and decoupled through the circuits labeled (2). Decoupling is actively controlled using the system bias current and trap circuits which produce high impedance in the coil loops when PIN diodes are forward biased. If bias current is not available due to a system failure, the traps will be activated passively by the RF power during transmission. These trap circuits ensure safe artifact free operation. Phase shift circuits (labeled {3}) are employed to improve inductive isolation using the system preamplifier impedance mismatch. Item 4 is the system interface which defines the pins associated with each coil channel.


System Bendix Connector
4
3
3
3
3
Connector
1 Combiner
2 Tuning, Matching, and Decoupling Circuit
3 Phase Shifting Circuit
4 (1.5T)
Pin Row 5 6 7 8 9 10 11 12
A GND GND GND GND
B S3 S4 S5 S6
4 Coax Cables
4 (1.0T)
Pin Row 5 6 7 8 9 10 11 12
A S3 GND S4 GND S5 GND S6 GND
B
S3
S4
S5
S6
1
2
2
2
2
2
U/L Saddle
Upper Neck
Lower Neck 1
Lower Neck 2
R/L Saddle
Coil

SECTION 2 - SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None

2-1-2 Installing the Coil

The names for this coil are: NVHEAD, NVARRAY, and NVFASTHD.

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; System Level Procedures; Software Utilities.

If the coil does not exist in Coil Config File refer to the Adding New Coils to Config File Manager procedure and use the coil configuration information in Section 7-3 of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.
2. Perform Section 3 - Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coils is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification.

SECTION 3 - FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

General tests: Be sure that you choose the correct coil, and activate the desired coils from the menu selection on the surface coil screen. The coil chosen must be NVARRAY.

Position the Array Coil at the head end of the patient cradle. Place the 16 cm diameter head phantom in the head portion of the HNC coil. Activate the NVARRAY coil from the soft key on your operator's console.

Scan using Auto Pre-scan. If Auto Pre-scan does not complete successfully, do not use the coil clinically. Upon completion of pre-scan, execute the scan. Window the scan as you normally would (window levels vary from system to system depending upon your hardware revision level). The scan should appear uniform.

If the image has signal voids, distortions, or black streaks, discontinue use of the coil. If the image is excessively noisy, repeat the scan using the GE 5 inch general purpose coil for 1.5T or alternate coil for 1.0T supplied with your system. If the 5 inch coil image or alternate coil image exhibits similar artifacts, the problem likely lies with the MRI system.

Reverse Polarity check: If the polarity on the system is uncertain, then use the following procedure to test coil operation. Position the Array Coil at the head end of the patient cradle. Place the 16 cm diameter head phantom in the head portion of the HNC coil. Activate the NVHEAD coil from the soft key on your operator's console. Choose a sequence used for SNR preventative maintenance with the GE Head coil, if possible.

Scan using Auto Pre-scan. If Auto Pre-scan does not complete successfully, due to low signal, manual pre-scan. A reverse polarity system would cause the SNR to be substantially lower than (less than 75%) the same sequence using the GE Head Coil. If this is observed the coil should be replaced with a reverse polarity version of the HNC Neurovascular coil.

3-3 External Cable Check

Refer to 3-4 Pin Diode Check.

3-4 PIN Diodes Check

A digital multimeter can be used to measure the PIN diode drop in each of the four elements. A single PIN diode drop of ~0.7 volts should be seen according to the block diagram at each of the inputs for S3, S4, and S5. The positive lead should go to the signal pins, S3-S6, and the negative lead should go to any of the ground pins, GND. Reversing the leads should read ~2.4 volts. For S6, which drives two loops, the typical forward drop is ~0.6 volts, while the reverse measurement is ~1.9 volts. If either reading is a short circuit, the diode(s) or the cable is shorted and a replacement cable should be used to retest the coil. If forwarding biasing the diode results in an open circuit or voltage over 1 volt, the diode(s) or cable is open and the coil must be replaced. The scanner will typically detect open and shorted channels and display a scan error.

PIN DIODE EXPECTED READINGS – TABLE 3-4

Positive Lead	Negative Lead	Reading
S3	GND	~0.7 Volts
S4	GND	~0.7 Volts
S5	GND	~0.7 Volts
S6	GND	~0.6 Volts
GND	S3	~2.4 Volts
GND	S4	~2.4 Volts
GND	S5	~2.4 Volts
GND	S6	~1.9 Volts

3-5 Mechanical Hardware Check

Not Applicable

3-6 Troubleshooting Tips

Not Available

SECTION 4 - MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The Open Neurovascular Array pad may be cleaned by wiping with a cloth dampened with a solution of 30% isopropyl alcohol and 70% tap water or a 10% Bleach solution.

4-2 Special Care Requirements

Not Applicable

SECTION 5 - REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 Disassembly of Coil

Not Applicable

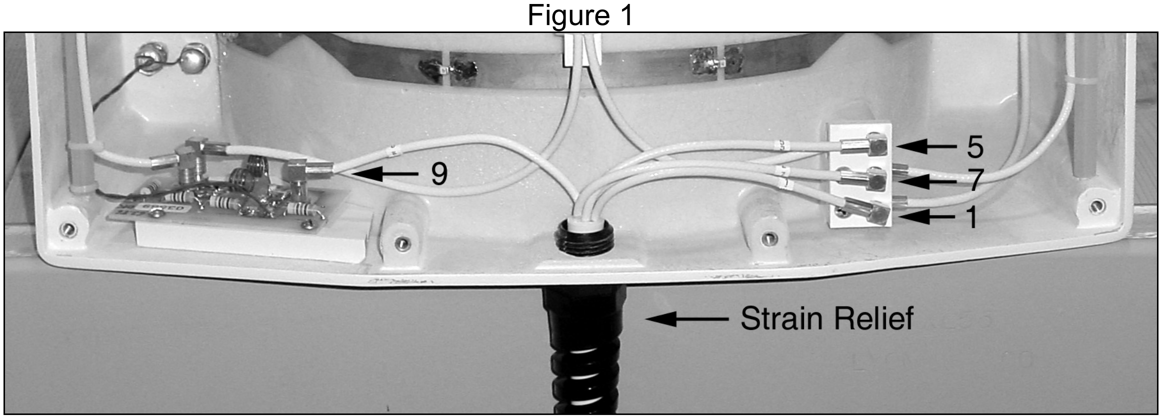
5-2 External Cable Replacement

To remove the external cable [for part number see Table 6-1 Field Replaceable Units]:

1. Remove the12 Phillips head screws located around the perimeter of the coil bottom. The coil bottom cover will now lift out of place.

2. Remove the bottom cover and place it in a safe location. Figure 1 is the bottom view of the coil with the cover removed.

Figure 1



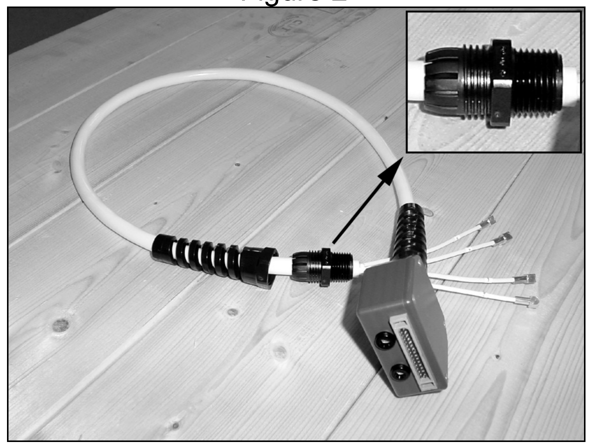
Coil Bottom View with Cover Removed

The coil cable assembly contains 4 individual coaxial cables as seen in Figure 1. Each individual coaxial cable is tagged for identification [1,5,7,9].

3. Disconnect each individual coaxial cable by pulling the connector out and away from the connector block or circuit board.

4. Group the individual coaxial cables and, using an adjustable open end wrench, loosen the strain relief plastic nut closest to the coil cover [counterclockwise] and remove the coil cable assembly.

Figure 2



Coil Cable Assembly

The Coil Cable Assembly uses a threaded plastic piece [Figure 2 Inset] that attaches the cable assembly to the coil cover **and** tightens the strain relief around the cable assembly.

The threads on the right, and the plastic nut, attach the cable assembly to the cover. The threads on the left tighten the strain relief around the cable assembly when the strain relief nut is tightened.

The steps for re-assembly using the replacement coil cable are as Follows:

1. One by one, thread the individual coaxial cables through the threaded hole in the coil cover.

2. Grouping the individual coaxial cables, screw the coil cable assembly plastic connector into the threaded hole in the coil cover. Snug the plastic nut down using an open end adjustable wrench.

3. Loosen the strain relief nut enough to allow the coil cable to rotate in the strain relief. Orient the coil cable such that, when the coil is laying with the bottom cover [and 12 screws] facing up, the system connector cover [and 4 screws] is also facing up. Snug down the strain relief nut using an open end adjustable wrench.

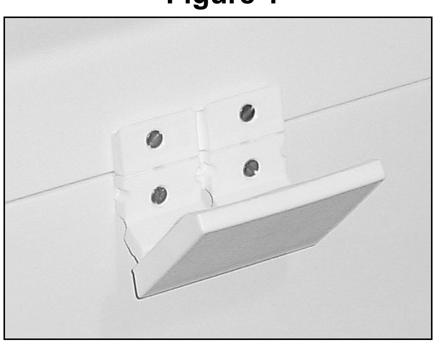
4. Connect the 4 individual coaxial cables using the cable tags and identifiers in Figure 1.

5. Using a Phillips head screw driver, replace the coil cover and 12 Philips head screws.

Verify coil operation according to Section 3-2, Coil Imaging Performance Verification.

5-3 Mechanical Hardware Replacement

Figure 1



Top Cover Latch

To replace an Open Neurovascular Array top cover latch, begin by removing the four 6-32 flat head machine screws that attach it to the coil cover. The latch will immediately separate from the coil cover.

Reverse the process to install the replacement latch. Note that the beveled head of the 6-32 screw, and the countersunk hole in the latch make it unnecessary to adjust the latch. Make sure to install all four screws loosely and then tighten all screws for a correct installation.

SECTION 6 - RENEWAL PARTS

6-1 Field Replaceable Units

1.0T FIELD REPLACEABLE UNITS LIST – TABLE 6-1-1

Description	GE Part #	MRIDC Part #	Qty
Latches	2225476-6	900214	1
Cable only	2225477-3	100015	1
Coil with Cable	2293669-2	101423	1

1.5T FIELD REPLACEABLE UNITS LIST – TABLE 6-1-2

Description	GE Part #	MRIDC Part #	Qty
Latches	2225476-6	900214	1
Cable only	2225476-3	100014	1
Coil with Cable	2293668-2	101413	1

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	GE Part #	MRIDC Part #	Qty
Wedge Pad	E8800NB	101407	1
Pad	E8800NC	101408	1

NOTE: Pads are the same for 1.0T and 1.5T.

SECTION 7 - APPENDIX

7-1 SNR Data Sheet

Not Applicable

7-2 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation

7-3 Coil Configuration

Parameter	1.5T	1.5T	1.5T	1.0T	1.0T
Coil Name	NVHEAD	NVARRAY	NVFASTHD	NVHEAD	NVARRAY
Coil Type	3	3	3	3	3
Extremity Coil	no	no	no	no	no
Cable Loss	1.05	1.05	1.05	1.24	1.24
Coil Loss (BRM/CRM)	1.72/0.952	1.72/0.952	1.72/0.952	0.549	0.549
Recon Scale Factor	1	1	1	1	1
Linear vrs Quadrature	1	1	1	1	1
Multiple Receiver Coil?	yes	yes	yes	yes	yes
Number of Receivers	1	4	1	1	4
Starting Receiver ID	0	0	0	0	0
Ending Receiver ID	0	3	0	0	3
Multi-Coil Port Enable	4	6	4	4	6
Multi-Coil Part Error Enable	4	6	4	4	6
Additional Transmit Attenuation	0	0	0	0	0
Number of Fast Receivers	1	0	1	0	0
Starting Fast Receiver ID	4	4	4	4	4
Ending Fast Receiver ID	4	4	4	4	4
Start TA Value	90	90	90	90	90
Start RG Value	12	12	12	12	12
Multi-Coil Recon Enable	0	0	0	0	0
Head Default Freq. Direction	1	1	1	1	1
SCIC Axial	0	0	0	0	0
SCIC Saggital	0	0	0	0	0
SCIC Coronal	0	0	0	0	0
Auto Shim Receiver	-1	-1	-1	-1	-1

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change	DCN
A	04-20-01	L. Hyler	First Issue	